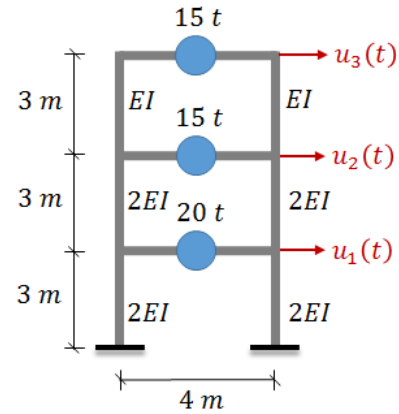


CE586 EARTHQUAKE ENGINEERING
2025-2026 SPRING SEMESTER
ASSIGNMENT #4 (DUE JUNE 12)

Q1. Consider the 3-story shear frame building model shown in the figure. It is assumed that the columns are massless and their flexural rigidities are as provided in the figure. There exists a lumped mass at each floor as shown. Neglect axial deformations in all elements. Obtain the differential equation of motion of this idealized structure for the lateral DOFs (i.e. u_i) only, subjected to a horizontal earthquake ground motion $\ddot{u}_g(t)$ by considering the following two cases separately:

- beams are flexurally rigid,
- beams on each story have flexural stiffness EI .



Consider $EI = 3600 \text{ kNm}^2$. Use Rayleigh damping with 5% damping ratio for modes 1 and 2. Employ “static condensation technique” whenever necessary. Compare and discuss the results briefly.

Q2. For the 3-story frame in Q1, carry out eigenvalue analysis for the case of no damping and flexurally rigid beams. Calculate eigenvalues and eigenvectors.

Q3. For the 3-story frame in Q1, answer the following questions (use the eigenvalue analysis results that you obtained in Q2):

- a) Calculate the modal properties of the frame, i.e. L_n , M_n , Γ_n , where $n=1-3$.
- b) Calculate and plot displacement history of the equivalent SDOF system for each uncoupled mode (D_n , $n=1-3$) subjected to the ground motion record which was already assigned to you in your third homework. Also record the maximum values and when they occur. Discuss the results.
- c) Calculate and plot the floor displacement response history $u_i(t)$, $i=1,3$. What are the maximum drifts for the 1st, 2nd and 3rd stories and when do they occur?
- d) Calculate and plot the story shear response history $V_i(t)$, $i=1-3$. Record the maximum values and when they occur.
- e) Calculate and plot the base overturning moment history $M_b(t)$. Record the maximum value and when it occurs.
- f) Calculate the effective modal masses and heights M_n^* and h_n^* , where $n=1-3$. Find the base shear and base overturning moment responses by using these effective modal parameters and compare your results with the ones in parts (d) and (e). Discuss the results.

Note: index “ n ” has been used for mode number whereas index “ i ” has been used for story number.

Q4. Consider the same frame building model given in Q1.

a) Construct the elastic response spectrum (with 5% damping ratio) by using the horizontal ground motion record assigned to you in the third homework.

b) Through the Response Spectrum Analysis (RSA), compute the maximum values for the following response quantities by considering all combination rules (ABSSUM, SRSS, CQC):

- floor displacements ($u_i, i=1,3$)
- story shears ($V_i, i=1-3$)
- base overturning moment (M_b).

c) After gathering all the above results in a single table together with the ones obtained in Q3 by using the Response History Analysis (RHA), comment on the accuracy of the employed procedures.

d) Obtain the maximum base shear and overturning moment by using the Equivalent Static Lateral Force Procedure (ESLFP) and considering the design spectrum (5% damping ratio) given below.

e) Compare the results with the ones obtained from RHA and RSA and discuss briefly.

